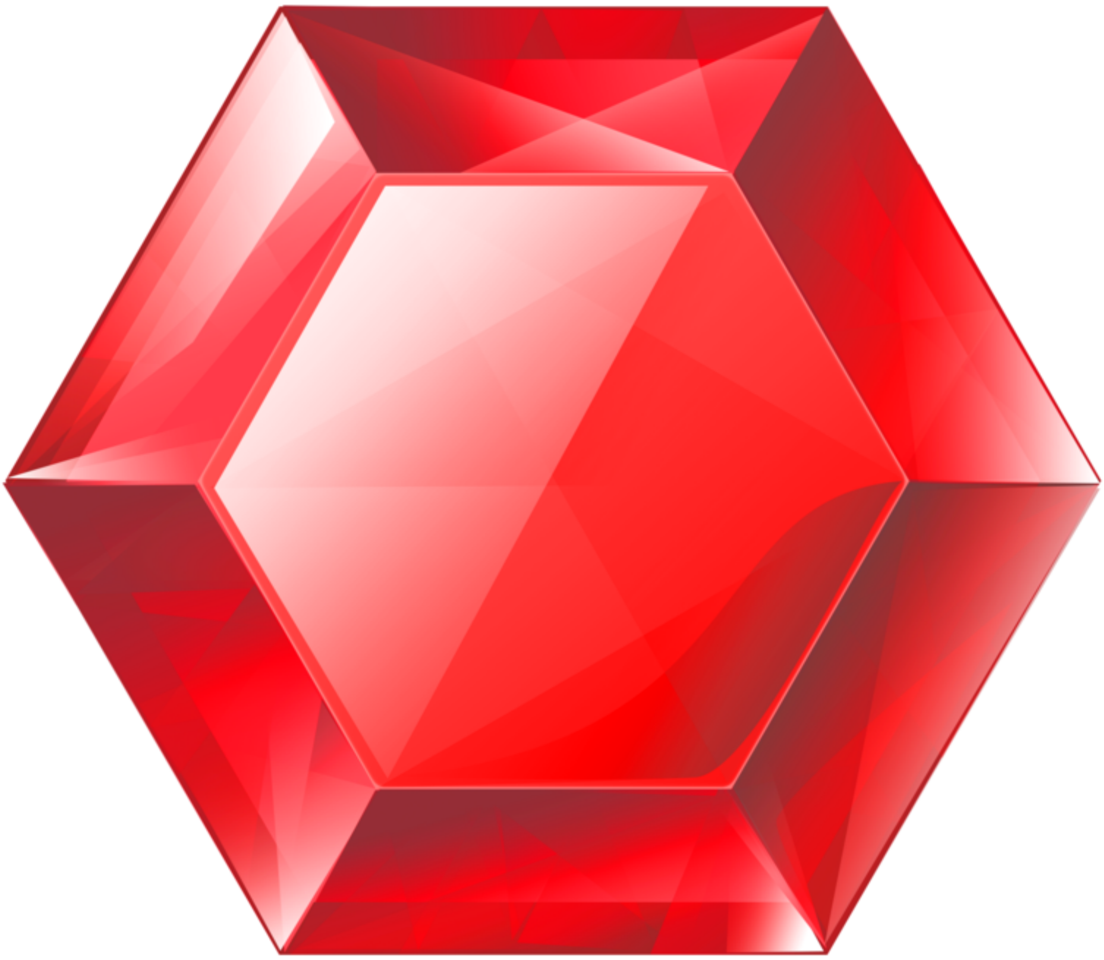
gemstone-rs Feature Map

Crates, examples, docs, and gemstone-py parity references



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gemstone-rs Feature Map

This map is the Rust equivalent of `gemstone-examples plan3-map` in `gemstone-py`: it ties each feature stream to the crates, examples, docs, and Python reference point that inspired it.

Run it from the CLI:

```
gemstone-rs hello
gemstone-rs compare gemstone-py
gemstone-rs compare gemstone-py --gaps
gemstone-rs examples map
gemstone-rs examples map --json
gemstone-rs examples scaffold quickstart ./gemstone-rs-quickstart
gemstone-rs examples scaffold codegen_workflow ./gemstone-rs-codegen-workflow
gemstone-rs examples scaffold generated_wrapper_app ./gemstone-rs-generated-wrapper
gemstone-rs examples scaffold axum_service ./gemstone-rs-axum-service
```

`hello` is the no-live sanity check that mirrors `gemstone-examples hello`. `compare gemstone-py` prints a compact version of the Python/Rust comparison guide. `compare gemstone-py --gaps` prints the remaining parity gaps with next actions and verification commands. `examples scaffold` creates standalone Cargo projects from installed templates for quickstart, browser, BridgeRoot mapping, derive mapping, generated wrappers, codegen, standard HTTP, and Axum workflows. JSON output is intended for CI, docs checks, and editor tooling.

Streams

Stream	Feature	Rust surface	Examples	Docs	gemstone-
1	Runtime and GCI loading	`gemstone-gci`, `gemstone-rs::Config`	`hello`, `hello_gemstone`, `quickstart`	`docs/setup-guide.md`, `docs/performance-safety.md`	`gemstone_` native backe
2	Safe sessions and transactions	`gemstone-rs::Session`, `TransactionGuard`	`quickstart`, `transactions`, `live_smoke_cookbook`	`docs/user-manual.md`, `docs/cookbook.md`	`GemStoneS` `SessionFa` transaction

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3	Browser and inspection	`gemstone-rs::browser`, CLI `browse`, `gemstone-rs-explorer`	`browser`, `tooling/cli-browser-walkthrough.md`	`docs/user-manual.md`, `docs/explorer.md`	`gemstone_` `python-ge` `database-e`
4	OOP and value handling	`Oop`, `Value`, export-set helpers	`oop_values`	`docs/user-manual.md`, `docs/performance-safety.md`	Managed O typed access
5	BridgeRoot object mapping	`gemstone-rs::bridge`, `gemstone-rs-macros`	`bridge_root_mapping`, `derive_mapping`, `generated_mapping_app`	`docs/object-mapping.md`, `docs/cookbook.md`	`Smalltalk` `Persisten` examples
6	Typed codegen	`gemstone-rs::codegen`, CLI `codegen` and `profile`	`codegen_preview`, `codegen_workflow`, `generated_wrapper_app`	`docs/codegen.md`, `docs/profile-schema.md`	`gemstone_` `typed_acc` `codegen_de`
7	Explorer workflow	`gemstone-rs-explorer`	`tooling/explorer.md`	`docs/explorer.md`, `docs/screenshots.md`	`python-ge` `database-e`
8	VS Code workbench	`vscode-gemstone-rs-workbench`	`tooling/vscode-workbench.md`	`docs/vscode-workbench.md`	`gemstone-`

Stream	Feature	Rust surface	Examples	Docs	gemstone-
9	Rust web services	`gemstone-rs` examples plus installed Axum scaffold and planned adapters	`http_service`, `axum-service/README.md`, `examples scaffold axum_service`	`docs/examples-guide.md`, `docs/cookbook.md`	FastAPI, Lit examples
10	Release and verification	`scripts`, `Makefile`, GitHub Actions	Release verification commands	`docs/release-checklist.md`	PyPI/TestPy VSIX releas
11	Shared native core	`gemstone-gci`, `gemstone-rs`, future `gemstone-py-native` wrapper	Shared-core integration plan	`docs/shared-core-integration.md`	`gemstone-`

What This Says Compared With gemstone-py

`gemstone-rs` is strongest where Rust is naturally valuable: direct native GemStone access, explicit OOP/value handling, typed generated wrappers, BridgeRoot mapping, and CLI/explorer tooling that can run without Python in the process.

`gemstone-py` remains stronger where Python already has mature product shape: checked web framework adapters, async examples, the database explorer, package extras, and broader release surfaces.

The projects should converge by sharing the Rust native core underneath `gemstone-py-native`, while keeping the Python and Rust APIs idiomatic for their respective users.

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